1. Abstract

This report outlines three key areas where Westlake Financial can improve operational efficiencies using Artificial Intelligence (AI): personalised product recommendations, credit risk assessment, and payment fraud detection. For each, it reviews models, data requirements, an implementation approach, and associated risks.

2. Introduction

The UK Financial Services (FS) sector has seen the rise of FinTech start-ups. Neobanks like Chase, Monzo, and Starling Bank exemplify the digital-first approach; for example, Monzo added 2.4 million customers in 2025, bringing its total customer base to over 12 million (Monzo, 2025, pp. 10-13). These FinTechs have seamlessly embedded AI technologies into their operational landscapes, improving operational efficiencies, reducing costs, and supporting regulatory responsibilities (Douglas, Barberis and Buckley, 2017), which is also the key aim of Westlake Financial.

3. Three Areas for AI Implementation

3.1. Personalised Product Recommendations

Westlake Financial can use AI to identify customer financial needs and deliver tailored product recommendations. AI models can predict the most relevant financial products for each individual. This enables the company to generate additional revenue through targeted cross-selling while improving customer satisfaction.

K-means clustering, an unsupervised learning algorithm, is one AI model that can be used in this area. This algorithm requires numerical data; for example, customer income may be used to sort customers into different income-based tiers that determine product recommendations. However, K-means does not consider explicit customer preferences, resulting in more generalised recommendations. K-prototypes clustering, an unsupervised machine learning (ML) algorithm, is a suitable alternative, as it uses both numerical and categorical data—such as customer preferences—allowing for individualised product recommendations.

Several banks have already adopted similar AI solutions. For example, JP Morgan Chase uses AI to deliver personalised product recommendations based on customer transactions, interactions, and financial goals (Tulsi, 2024). In this way, Westlake Financial should consider adopting this approach; however, it may be more appropriate to implement less complex models than JP Morgan Chase initially.

3.2. Credit Risk Assessment

Credit risk analysis, scoring, and loan approval are areas where AI can improve operational efficiency and increase revenue by reducing reliance on manual workflows, which carry overhead costs, while enabling faster and more consistent decisions.

Logistic Regression (LR) has traditionally dominated credit risk assessment; however, academic studies indicate that ensemble tree-based methods—including Decision Trees (DTs) and Random Forests (RFs)—produce more robust and accurate models. Lessmann *et al.* (2015) demonstrate that these models have started to establish themselves as the industry benchmark by outperforming LR-based models. The change is evident among FinTechs; Zopa, for example, has adopted non-LR-based credit risk models (Fanning, 2019).

DTs are easily explainable due to their rule-based structure, but are prone to overfitting, often resulting in inaccurate predictions. RFs address this by aggregating multiple DTs, reducing variance, and improving generalisation at the cost of explainability. Westlake Financial needs to consider these approaches if the company wants to compete with the Neobanks; however, analysis, data preprocessing, and experimentation are essential to balance predictive performance with regulatory obligations. This ensures any implemented solution is both effective and compliant.

3.3. Payment Fraud Detection

The UK FS sector reported £1.17 billion in fraud losses in 2024, up by 0.4% from 2023 (UK Finance, 2025). Automated fraud detection presents Westlake Financial with an opportunity to mitigate further losses.

As with credit risk assessment models, financial institutions typically use rule-based supervised learning approaches for fraud detection (Uddin, 2025). Gradient Boosting, an ensemble supervised learning method, is particularly well-suited to payment fraud detection. It handles class imbalance (i.e., fraudulent transactions), captures complex non-linear patterns, processes high transaction volumes, and produces interpretable outputs.

Recent studies support the effectiveness of gradient boosting models. Detthamrong *et al.* (2024) evaluate ML algorithms such as LightGBM, XGBoost, and CatBoost, concluding that CatBoost is the most accurate for identifying fraudulent transactions. However, beyond accuracy, minimising latency must also be considered, as payments occur 24/7, which means models require speed and efficiency to prevent losses while simultaneously balancing customer expectations.

Barclays exemplifies successful implementation, having scaled its fraud system to ensure accurate detection while reducing latency (Aerospike, no date). However, as a start-up with comparatively lower transaction volumes, Westlake Financial may find an RF model a more appropriate and cost-effective starting point.

4. Ethical, Legal, and Professional Issues

4.1. Algorithmic Bias

In considering the solution areas outlined in Section 3, biased training datasets risk unfair customer treatment and discriminatory outcomes such as declined loan applications due to a customer's race or gender, which goes against the principles defined in the Equality Act 2010 that prohibits discrimination on the basis of protected characteristics. One such organisation that has addressed this challenge is Zest AI, which reports that its ML technology has improved loan approval rates for persons of colour and women (Zest AI, 2025).

4.2. Transparency and Explainability

AI systems are often perceived as black boxes, which is a major concern for financial institutions. Banks must explain loan decisions for algorithmic bias purposes while also guarding against data drift—that is, a change in data—over time. As the Bank of

England (2022) notes, “the use of complex and/or opaque AI systems to model probability of default and loss given default in IRB credit models could lead to firms having incorrect levels of regulatory capital.” Consequently, Explainable AI (XAI) has emerged, with providers like Temenos (2024) integrating XAI into their Core Banking Software. Westlake Financial should adopt XAI when implementing its AI solutions.

4.3. Workforce Displacement and Hallucinations

AI deployment may lead to workforce displacement, requiring reskilling as new roles emerge in model monitoring and validation. Moreover, AI systems can often produce inaccurate outputs termed hallucinations, as demonstrated by Deloitte whose AI-generated content contained errors (Paoli, 2025). Such failures create operational and reputational risks, particularly in high-stakes financial decision-making. As a result, Westlake Financial must employ human-in-the-loop mechanisms to mitigate these risks. This may include having operations staff monitoring AI-generated credit decisions, reviewing fraudulent transactions, and maintaining data and models to prevent model drift.

4.4. Data Governance and Privacy Concerns

AI systems depend on data; however, an organisation’s use of personal data is governed by GDPR and the Data Protection Act 2018. These regulations restrict the unlimited use of personal data, outlining that organisations must ensure explicit consent (i.e., opt-in/opt-out) or have substantiated legal reasons (e.g., financial crime monitoring) for using their customer data. This applies to the use of customer data across the three AI systems outlined in Section 3.

5. Deployment Approach and Data Requirements

5.1. Data Quality and Feature Selection

Data quality and feature engineering are fundamental prerequisites for effective AI models (Duboue, 2020). A clear understanding of the desired outcome is essential when sourcing and preprocessing data, as it directly influences feature selection, data transformation, the handling of biases, noise, and missing values.

As a start-up, Westlake Financial lacks in-house data and must source datasets externally (e.g., Kaggle) or generate synthetic data with GenAI tools such as Claude AI or ChatGPT (see Appendix 1). All data should be validated prior to AI model training, with any limitations documented as evidenced by Flonder, Donath, and Neamtu (2024), whose card fraud detection model relied on synthetic data that may not accurately represent real-world conditions.

5.2. Specific Data Requirements by Model

5.2.1 Personalised Product Recommendation

These models require access to an up-to-date product catalogue which includes current accounts, savings accounts, and unsecured lending products. This data should also include information relating to credit card balance transfer incentives, cash back rewards, and bonus interest rates on savings products. The model requires access to customer-specific behavioural data such as transaction history (see Section 5.2.3), expressed customer preferences, and financial data (see Section 5.2.2) to determine customer eligibility, ensuring product suggestions are appropriate given each customer's financial profile.

5.2.2 Credit Risk Assessment Data Requirements

Credit risk assessment models must incorporate internal policies and business rules to ensure alignment with Westlake Financial's credit risk strategy. Key data sources needed include borrower identity and demographic information, income and expenditure details, credit history from internal systems and from credit reference agencies such as Experian and Equifax. Behavioural and transactional data (see Section 5.2.3) are also required. Open Banking data may also be available.

5.2.3 Fraud Detection Data Requirements

Real-time fraud detection models require large volumes of transactional data to identify patterns and anomalies. Key data points, or features, in the transactional data should contain labels such as Transaction Amount, Type, Geolocation, Merchant ID, Transaction Time, and Payment Channel to derive patterns. Additionally, Customer Profile data is required, which includes datapoints such as the customer's

age and occupation, to establish baseline behavioural patterns and identify deviations that may indicate fraud.

5.3. Implementation Approach

Westlake Financial should adopt the industry standard CRISP-DM process model across its six phases (see Figure 1) when implementing AI solutions.

Phase	Short description
Business Understanding	The business situation should be assessed to get an overview of the available and required resources. The determination of the data mining goal is one of the most important aspect in this phase. First the data mining type should be explained (e. g. classification) and the data mining success criteria (like precision). A compulsory project plan should be created.
Data understanding	Collecting data from data sources, exploring and describing it and checking the data quality are essential tasks in this phase. To make it more concrete, the user guide describe the data description task with using statistical analysis and determining attributes and their collations.
Data preparation	Data selection should be conducted by defining inclusion and exclusion criteria. Bad data quality can be handled by cleaning data. Dependent on the used model (defined in the first phase) derived attributes have to be constructed. For all these steps different methods are possible and are model dependent.
Modeling	The data modelling phase consists of selecting the modeling technique, building the test case and the model. All data mining techniques can be used. In general, the choice is depending on the business problem and the data. More important is, how to explain the choice. For building the model, specific parameters have to be set. For assessing the model it is appropriate to evaluate the model against evaluation criteria and select the best ones.
Evaluation	In the evaluation phase the results are checked against the defined business objectives. Therefore, the results have to be interpreted and further actions have to be defined. Another point is, that the process should be reviewed in general.
Deployment	The deployment phase is described generally in the user guide. It could be a final report or a software component. The user guide describes that the deployment phase consists of planning the deployment, monitoring and maintenance.

Figure 1: CRISP-DM Approach (Schröer, Kruse and Gómez, 2021, p. 527)

Sections 2 and 3 of this report form the basis for CRISP-DM's Business Understanding phase, while Section 5 outlines the types of data and feature selection required to build suitable AI models. The organisation must continue to follow this process when commencing the implementation stage of the project, ensuring adequate data preparation and preprocessing actions are taken before commencing with model creation and evaluation.

6. Conclusion

Westlake Financial can use AI to improve operational efficiency and remain competitive in the digital financial landscape. As a start-up, it should adopt a phased implementation approach, beginning with simpler, cost-effective models before scaling complexity as data and transaction volumes grow. Adherence to the CRISP-DM framework, rigorous data governance, and regulatory compliance will ensure any deployed solution is both effective and responsible.

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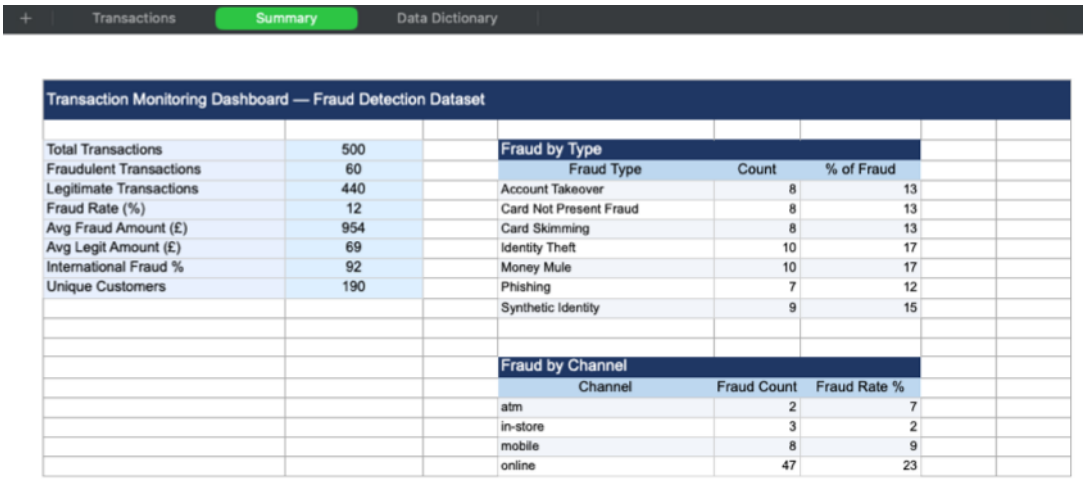
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Appendices

1. Claude AI was used to generate a sample dataset that could be used to train a fraud detection model in Figure 1 below:

   The above dataset was generated using a straightforward prompt. More precise outputs can be achieved by incorporating more defined parameters. This highlights the importance of iterative experimentation during model development and synthetic dataset generation.

   Prompt and dataset: https://claude.ai/chat/fdd677a4-4a3f-4a49-9949-60b353e8bdc7



Transactions | Summary | Data Dictionary

Transaction Monitoring Dashboard — Fraud Detection Dataset

Total Transactions	500
Fraudulent Transactions	60
Legitimate Transactions	440
Fraud Rate (%)	12
Avg Fraud Amount (£)	954
Avg Legit Amount (£)	69
International Fraud %	92
Unique Customers	190

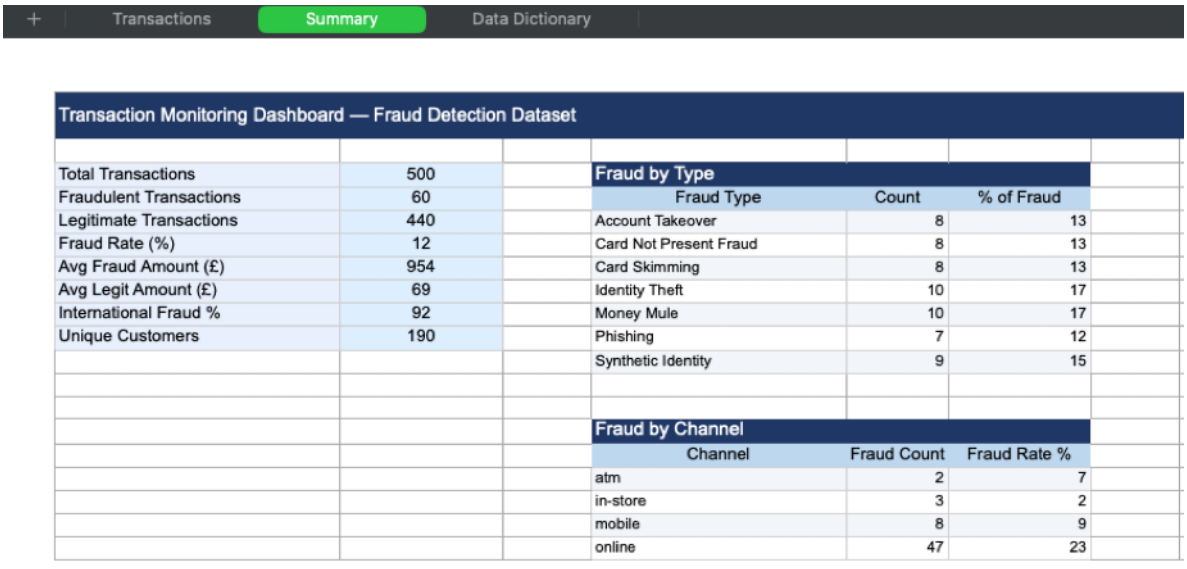
Fraud by Type

Fraud Type	Count	% of Fraud
Account Takeover	8	13
Card Not Present Fraud	8	13
Card Skimming	8	13
Identity Theft	10	17
Money Mule	10	17
Phishing	7	12
Synthetic Identity	9	15

Fraud by Channel

Channel	Fraud Count	Fraud Rate %
atm	2	7
in-store	3	2
mobile	8	9
online	47	23

Figure 2: GenAI produced Synthetic Dataset Tab A (Anthropic Claude, 2026)



Transactions | Summary | Data Dictionary

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Phishing	7	12
Synthetic Identity	9	15

Fraud by Channel

Channel	Fraud Count	Fraud Rate %
atm	2	7
in-store	3	2
mobile	8	9
online	47	23

Figure 3: GenAI produced Synthetic Dataset Tab B (Anthropic Claude, 2026)



Figure 3: GenAI produced Synthetic Dataset Tab C (Anthropic Claude, 2026)